Portfolio Construction for Pension Funds

From Unstable Estimates to Dynamic Decisions

Group 4

School of Mathematics and Statistics
University of Sydney

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Introduction

Research question

How can historical market data be used to construct reliable strategic asset allocations for pension funds, and can tactical reallocation and wealth-dependent multiperiod decisions generate additional value?

Our approach

- Construct strategic portfolios.
- Assess sensitivity and out-of-sample performance.
- Compare tactical and dynamic allocation strategies.

Data and Monthly Returns

- **Data coverage**
  Nine Bloomberg indices representing nine asset classes.
- **Common sample**
  248 monthly returns from January 2006 to August 2026, with no missing observations in the common sample.
- **Why monthly?**
  95.4% of HEDGNAV's weekday observations are unchanged. Monthly data better reflect its update frequency and reduce the impact of different market closing times.

From Returns to Model Inputs

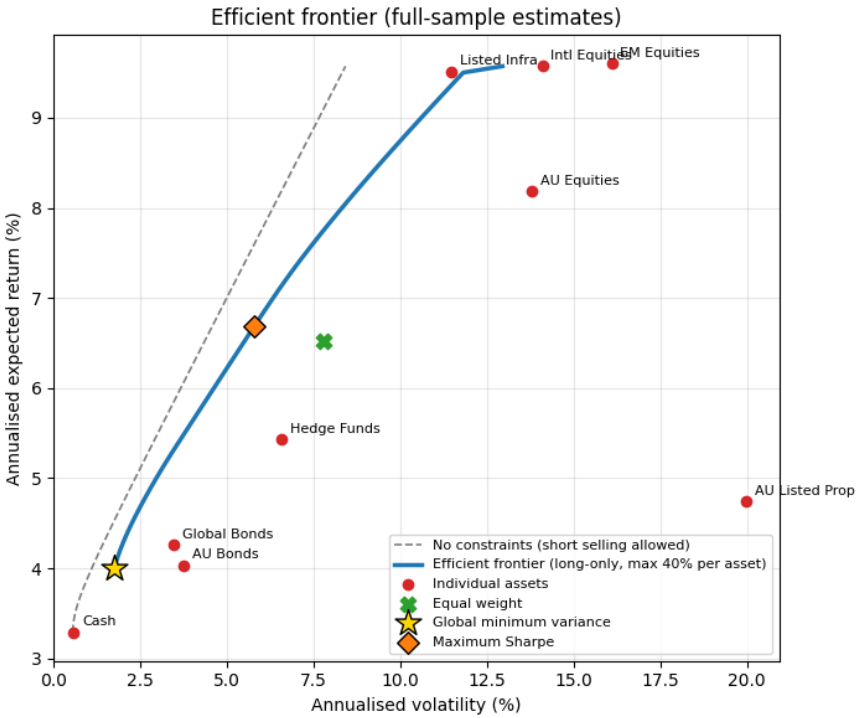
- **Desmoothing**
  Adjust smoothed returns to better capture the underlying risk of hedge funds.

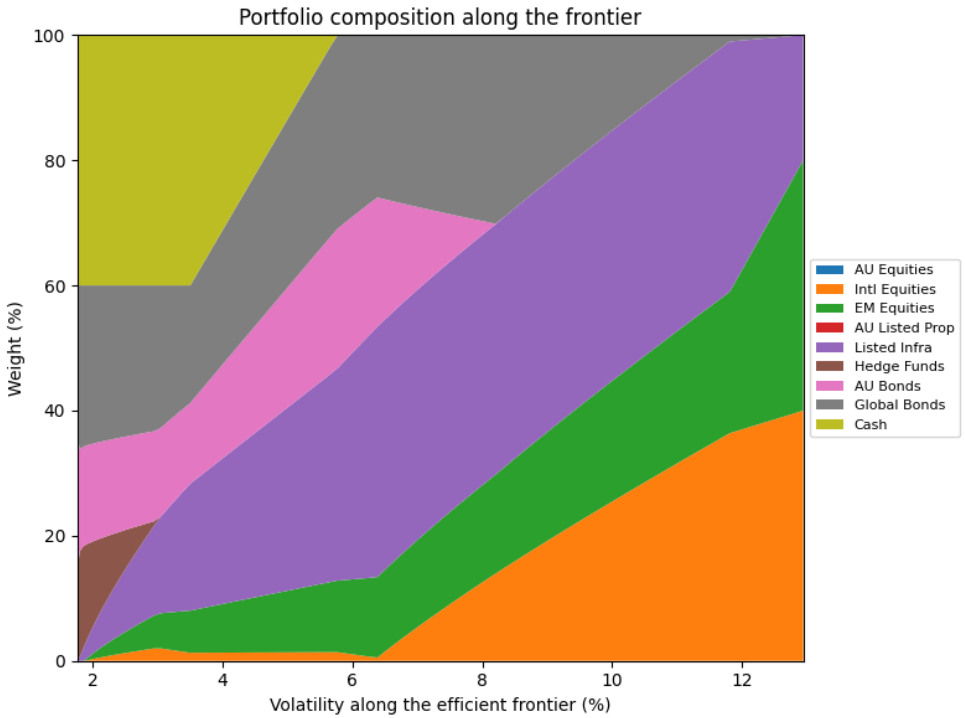
- **Annualisation**
  Convert monthly mean returns and covariances into annual inputs, assuming negligible monthly autocorrelation.

- **Portfolio inputs**
  Estimate the expected return vector $\mathbf{m}$ and covariance matrix $\mathbf{C}$ for portfolio optimisation.

Selected constraints

Chosen constraints: Fully invested, long-only, maximum 40% per asset.





Why? Long-only avoids leveraged long–short positions and keeps weights easy to interpret. The 40% cap limits concentration and requires at least three assets to have positive weights.

GMV: 4.00% expected return; 1.76% volatility.

Portfolio sensitivity to the estimation period

- $D = \frac{1}{2}\sum_i |x_i - x_i^{\text{full}}|$ measures the share of capital that would need to be reallocated to match the full-sample portfolio.
- We keep the constraints from the previous slide fixed. GMV weights shift little, while 6% target weights shift much more.

Alternative window	**GMV distance**	**6% target distance**
Last 10 years	14.0%	72.8%
Last 5 years	14.8%	70.7%

Why the difference matters

GMV uses covariance C alone; target-return weights also depend on estimated returns m, whose estimates vary across windows, so the weights shift more.

Strategic portfolio construction by growth target

Method: Construct three strategic asset allocations (SAAs) with growth exposure of 30%, 70%, or 90%, then minimise variance at each target.

SAA	Mix G/D	Return	Vol.	Sharpe
Defensive	30/70	5.0%	3.3%	0.519
Balanced	70/30	7.2%	7.6%	0.510
Aggressive	90/10	8.3%	10.1%	0.502

Interpretation: Higher growth raises expected return and volatility, while Sharpe ratios remain similar. The three targets offer different risk–return trade-offs, with no clear Sharpe-ratio winner.

Note: Growth/defensive classification follows the APRA framework; hedge funds are assumed to be 50/50. Annualised estimates; Sharpe uses mean cash return (3.29%) as the risk-free rate.

Numerical accuracy versus estimation uncertainty

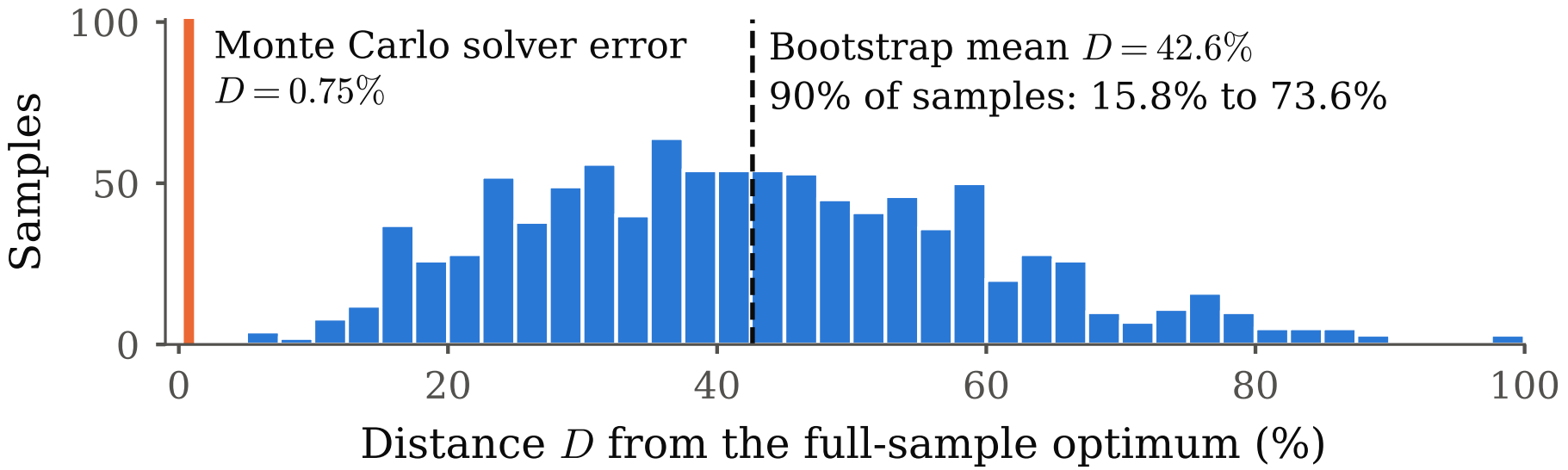
Risk preference: CARA utility, $\gamma = 10$ (close to maximum Sharpe).

Solving error: Monte Carlo

Same m and C, 256,000 simulated scenarios with antithetic variates.
Distance to the exact optimum: **0.75%**

Estimation error: bootstrap

Resample the 248 months 1,000 times, re-estimate m and C, re-solve.
Average distance: **42.6%**



Takeaway: estimation (mostly expected returns), not the solver, limits reliability.

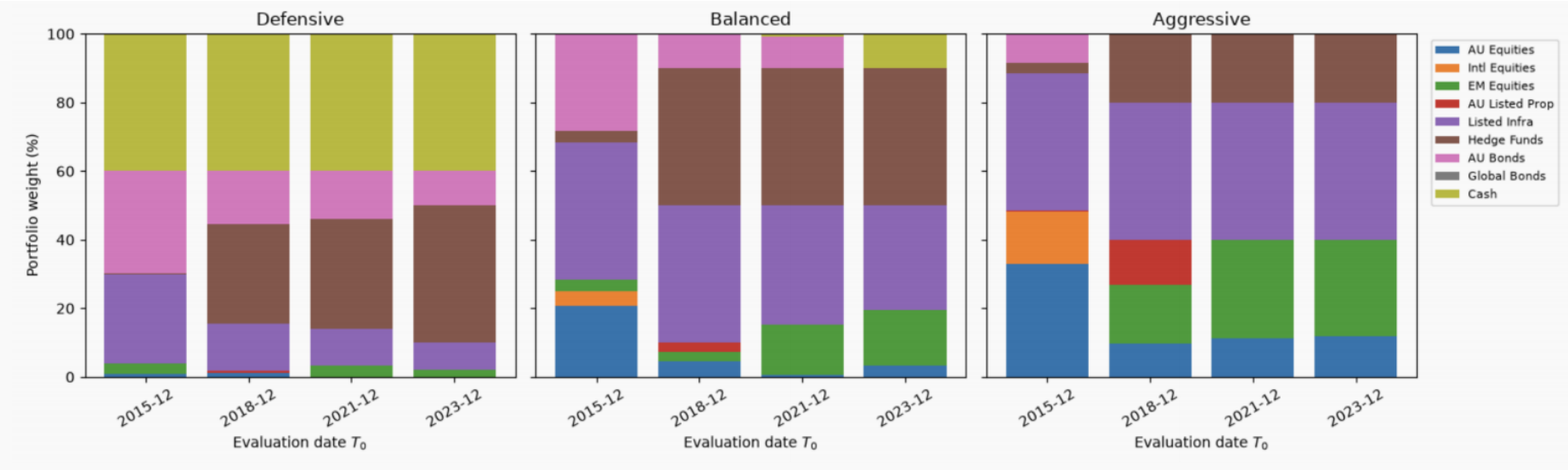
Out-of-sample design

Formation dates: Dec. 2015, 2018, 2021, 2023

Estimate: previous **120 months** $\rightarrow$ re-estimate m_{T_0} and C_{T_0}

Test: following **24 months** of unseen returns

Mandates fixed: Defensive 30%, Balanced 70%, Aggressive 90% growth

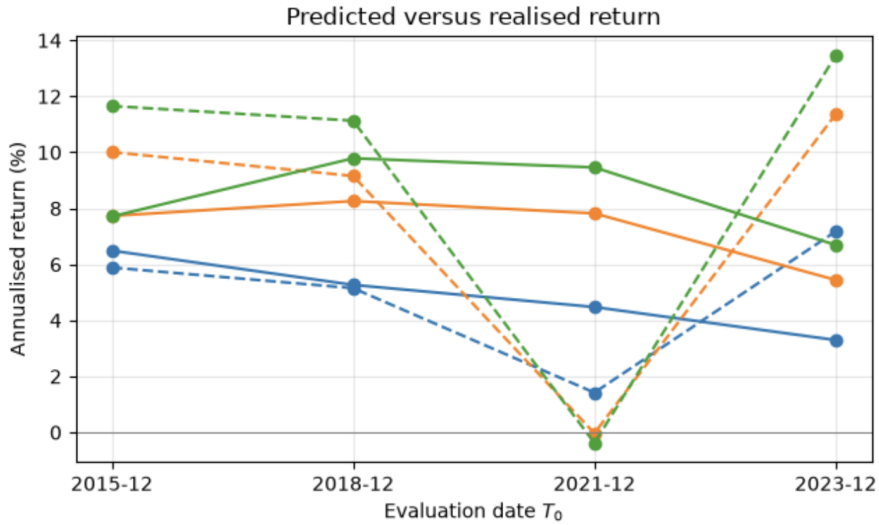


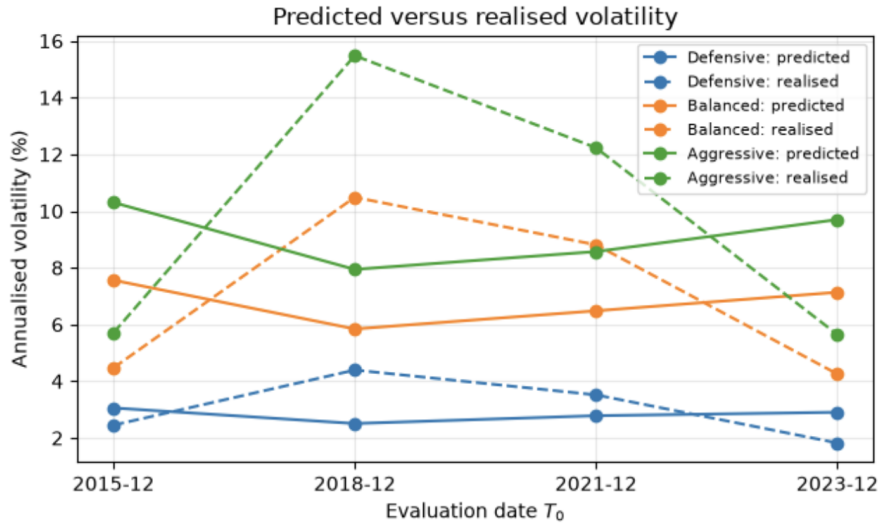
Same mandate + updated covariance $\Rightarrow$ different target weights

Target weights are restored monthly within each test; only information available at T_0 enters portfolio construction.

Stable allocations, different outcomes

The horizontal axis shows formation date T_0; realised performance covers the **following 24 months**.





Predicted and realised returns also diverge substantially across formation dates.

Balanced portfolio formed Dec. 2021: predicted return **7.82% p.a.**, but realised approximately **0% p.a.** over 2022–2023.

Finding

Historical estimates do not guarantee future performance.

Tactical Asset Allocation(TAA): performance

Design: Update estimates and weights monthly using a rolling 120-month window, with $\gamma = 10$ and 60–80% growth exposure.

Comparison: Fixed balanced SAA, with its target weights restored monthly.

Test period	TAA minus fixed SAA cumulative return (percentage points)
2016–2017	−0.54
2019–2020	−8.36
2022–2023	−6.64
2024–2025	−0.33

Each test covers 24 months. Returns exclude transaction costs.

Finding

Baseline TAA underperformed fixed SAA in all four tests. The largest shortfall was 8.36 percentage points in 2019–2020.

Tactical Asset Allocation(TAA): trading and limitations

Trading costs

- TAA turnover was approximately 6–16 times SAA turnover.
- Transaction costs widened the baseline shortfall.

Sensitivity

- A 60-month window improved cumulative returns in all four tests, but beat SAA in only two.
- Changing risk aversion also gave mixed results.

Comparison limitation

The objective and growth constraints also differ between TAA and SAA, so we cannot isolate the effect of updating estimates.

Takeaway

More flexible allocation showed no consistent benefit across these tests.

Dynamic allocation: wealth and time

TAA updates market estimates each month. Our dynamic policy keeps parameters fixed and lets $x_k = \pi_k(W_k)$ depend on wealth and time remaining.

Objective: maximise expected exponential utility of terminal wealth.

Bellman recursion

$$V_k(w) = \max_{x \in \mathcal{X}} \mathbb{E}\Big[V_{k+1}\Big(w(1 + x^\top R_{k+1}) \Big) \Big] .$$

$V_k(w)$: maximum expected terminal utility from wealth w at date k; terminal condition: $V_N(w) = -e^{-\gamma w}$.

Constraints: fully invested, long-only, 40% cap per asset. No growth restriction in the basic model.

Wealth dependence in the final period

For $w > 0$ and normal monthly returns, the final-period problem becomes

$$x^*_{N-1}(w) = \arg\max_{x \in \mathcal{X}} \left\{ x^\top m_\Delta - \frac{\gamma w}{2} x^\top C_\Delta x \right\}.$$

m_Δ and C_Δ: monthly return mean and covariance.

Effective risk aversion: γw
With parameters and constraints fixed, higher wealth does not raise optimal percentage volatility.

Earlier decisions require the full Bellman continuation value.

Dynamic allocation: method and result

- **Question:** Does adjusting the portfolio to current wealth improve investment outcomes?
- **Numerical validation:** Independent simulation gives CE 1.018082, compared with 1.017964 from backward induction. The difference is smaller than the Monte Carlo standard error.
- **Fair comparison:** Compare dynamic allocation with fixed mean–variance weights under the same model and constraints, using 50,000 independent paths.

Result

The certainty-equivalent gain is approximately -0.01 basis points of initial wealth. Its 95% Monte Carlo interval, $[-0.038,\ 0.018]$, includes zero: **no clear additional utility gain**.

Why? Wealth changes are small over three months, and portfolio limits restrict allocation changes.

Conclusion and limitations

- **Estimation is the main challenge.** Solving error 0.75% against estimation error 42.6%: read optimised weights as ranges, not targets.
- **What survives:** the broad structure, a growth–defensive split with infrastructure as the main growth holding and no listed property.
- **Extra flexibility added no consistent value.** TAA trailed the fixed SAA in all four tests while trading 6 to 16 times as much; the dynamic policy's CE gain (-0.01 bp) has an interval containing zero.

Limitations

About 20 years of data and four out-of-sample tests; iid normal returns; a three-month dynamic horizon with fixed parameters; month-end data assumed available immediately.

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Thank you!